

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science **ASSESSMENT**

TOOL 3 – Mock Interview

Course Code:	DSA0308	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	T.Megha Harsha	Reg. No.:	192424389
Section / Batch:	2024	Date:	

Code of Conduct

I T Megha Harsha(192424389) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: MEGHA

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5

Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

CHATGPT PROMPT: copy and paste in your chatgpt

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:

Language Generation and Discourse Analysis Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:

Ask the student to enter:

Student Name:

Register Number:

Step 2:

Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
 - 2 questions on Discourse and Coherence
 - 2 questions on Conversational Agents and Dialog Acts
 - 2 questions on Language Generation
 - 2 questions on Machine Translation
 - 2 analytical or scenario-based questions

Step 3:

Store every question and answer.

Step 4:

After Question 10, evaluate the student using the following rubric.

RUBRIC

Technical Knowledge (25) Level

4:

Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:

Shows good knowledge with minor gaps

Level 2:

Basic knowledge with noticeable errors

Level 1:

Lacks fundamental technical knowledge

Problem Solving (25) Level

4:

Solves problems logically and applies concepts effectively in real scenarios

Level 3:

Applies concepts with moderate accuracy

Level 2:

Limited problem-solving ability; requires guidance

Level 1:

Unable to solve or apply concepts

Analytical Thinking (20) Level

4:

Analyzes questions critically and provides well-structured logical answers

Level 3:

Provides reasonable analysis with some structure

Level 2:

Superficial or partially structured responses

Level 1:

No analytical thinking demonstrated

Communication (15) Level

4:

Communicates clearly, confidently and professionally using appropriate terminology Level 3:

Communicates well with minor hesitation

Level 2:

Communication lacks clarity or confidence

Level 1:

Unable to communicate effectively

Confidence (15) Level

4:

Displays high confidence, positive attitude and professional behavior throughout

Level 3:

Shows good confidence with minor issues

Level 2:

Low confidence and inconsistent behavior

Level 1:

Lacks confidence and professionalism

MARK CALCULATION

Level 4 = 100% of weightage

Level 3 = 75% of weightage

Level 2 = 50% of weightage

Level 1 = 25% of weightage

Technical Knowledge = 25 marks

Problem Solving = 25 marks

Analytical Thinking = 20 marks

Communication = 15 marks
Confidence = 15 marks

Calculate:

Final Score = Sum of all rubric scores

Step 5:

Generate the final report in the exact format below.

MOCK INTERVIEW ASSESSMENT REPORT

Student Name:

Register Number:

INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked: What is a dialog in NLP?

Student Response: A dialog is an exchange of information between two or more participants through language. In NLP, dialog systems are designed to understand user inputs and generate appropriate responses.

Q2:

Question Asked: What is a conversational agent?

Student Response: A conversational agent is a computer system that interacts with users using natural language. Examples include chatbots, virtual assistants, customer support bots, and voice assistants.

Q3:

Question Asked: What is machine translation?

Student Response: Machine Translation is the automatic translation of text or speech from one natural language into another. It uses linguistic rules, statistical models, neural networks, or combinations of these approaches.

Q4:

Question Asked: Analyze the reference in: "The dog chased the cat because it was afraid."

Student Response: The pronoun "it" is ambiguous because it may refer to either the dog or the cat. Semantic and contextual reasoning are required to determine which animal was afraid.

Q5:

Question Asked: Analyze the difference between local and global coherence.

Student Response: Local coherence refers to meaningful connections between nearby sentences. Global coherence refers to the overall logical consistency and topic development across the entire document.

Q6:

Question Asked: How can conversational agents be used in education?

Student Response: Educational conversational agents can answer student questions, provide explanations, conduct quizzes, and offer learning guidance. They can also adapt their responses according to student interactions.

Q7:

Question Asked: How can reference resolution improve search engines?

Student Response: Reference resolution helps search systems understand relationships between entities mentioned in a query or document. This improves retrieval accuracy by identifying what pronouns and referring expressions represent.

Q8:

Question Asked: How can machine translation be used in online learning platforms?

Student Response: Machine translation can translate learning materials, instructions, and student discussions into different languages. This makes educational content more accessible to learners worldwide.

Q9:

Question Asked: A sentence generator produces: "Yesterday market I went the." Which NLG stage failed?

Student Response: The surface realization stage has failed because the sentence has incorrect word order. The system must apply grammatical rules to generate the correct structure, such as "Yesterday, I went to the market."

Q10:

Question Asked: A chatbot conversation begins with a greeting, continues with information collection, and ends with confirmation. What does this demonstrate?

Student Response: This demonstrates discourse structure and dialog organization within a conversation. The dialog follows a meaningful sequence of interaction stages to complete a specific task.

RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	(15)	(15)	Communication Total	Confidence Final Mark	SubQ Level X	Student Avg Level X	Avg Level X
Level X	Level X	XX.X	XX.X						

OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

- Technical Knowledge Level
- Problem Solving Level
- Analytical Thinking Level
- Communication Level
- Confidence Level
- SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

- Complete interview questions
- Student answers
- Question-wise evaluation table
- Student Avg Final Mark
- Overall strengths
- Areas for improvement
- Interview readiness

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	3	3	3	2	3	14	69%
2	3	2	3	3	3	14	
3	2	3	3	2	3	13	
4	2	3	3	3	3	14	
5	3	3	2	3	3	14	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____